

```
In [20]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
In [3]: df=pd.read_csv("iris.csv")
df
```

```
Out[3]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

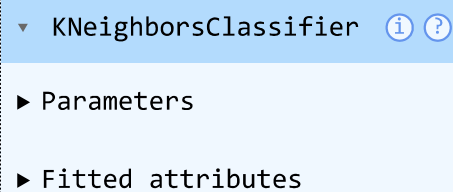
150 rows × 5 columns

```
In [4]: X=df[['sepal_length','sepal_width','petal_length','petal_width']]
Y=df['species']
```

```
In [5]: X_train, X_test, Y_train, Y_test=train_test_split(X,Y,test_size=0.2, random_state=42)
```

```
In [6]: scaler=StandardScaler()  
x_train=scaler.fit_transform(X_train)  
X_test=scaler.transform(X_test)
```

```
In [10]: k=5  
knn=KNeighborsClassifier()  
knn.fit(x_train,Y_train)
```

```
Out[10]:  KNeighborsClassifier ⓘ ?  
  ▸ Parameters  
  ▸ Fitted attributes
```

```
In [14]: Y_pred=knn.predict(X_test)
```

```
In [16]: accuracy=accuracy_score(Y_test,Y_pred)  
accuracy
```

```
Out[16]: 1.0
```

```
In [17]: conf_matrix=confusion_matrix(Y_test,Y_pred)  
conf_matrix
```

```
Out[17]: array([[10,  0,  0],  
                [ 0,  9,  0],  
                [ 0,  0, 11]])
```

```
In [ ]:
```